

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. Centres of the three circles $x^2 + y^2 - 4x - 6y - 14 = 0$, $x^2 + y^2 + 2x + 4y - 5 = 0$ and $x^2 + y^2 - 10x - 16y + 7 = 0$
 (A) are the vertices of a right triangle
 (B) the vertices of an isosceles triangle which is not regular
 (C) vertices of a regular triangle
 (D) are collinear

MCR001

2. $y - 1 = m_1(x - 3)$ and $y - 3 = m_2(x - 1)$ are two family of straight lines, right angled to each other. The locus of their point of intersection is
 (A) $x^2 + y^2 - 2x - 6y + 10 = 0$ (B) $x^2 + y^2 - 4x - 4y + 6 = 0$
 (C) $x^2 + y^2 - 2x - 6y + 6 = 0$ (D) $x^2 + y^2 - 4x - 4y - 6 = 0$

MCR002

3. Suppose that the equation of the circle having $(-3, 5)$ and $(5, -1)$ as end points of a diameter is $(x - a)^2 + (y - b)^2 = r^2$. Then $a + b + r$, ($r > 0$) is
 (A) 8 (B) 9 (C) 10 (D) 11

MCR003

4. The area of an equilateral triangle inscribed in the circle $x^2 + y^2 - 2x = 0$ is
 (A) $\frac{3\sqrt{3}}{4}$ (B) $\frac{3\sqrt{3}}{2}$ (C) $\frac{3\sqrt{3}}{8}$ (D) none

MCR004

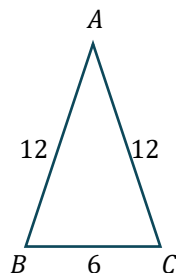
5. The smallest distance between the circle $(x - 5)^2 + (y + 3)^2 = 1$ and the line $5x + 12y - 4 = 0$, is
 (A) $1/13$ (B) $2/13$ (C) $3/15$ (D) $4/15$

MCR005

6. The equation of the image of the circle $x^2 + y^2 + 16x - 24y + 183 = 0$ by the line mirror $4x + 7y + 13 = 0$ is
 (A) $x^2 + y^2 + 32x - 4y + 235 = 0$ (B) $x^2 + y^2 + 32x + 4y - 235 = 0$
 (C) $x^2 + y^2 + 32x - 4y - 235 = 0$ (D) $x^2 + y^2 + 32x + 4y + 235 = 0$

MCR006

7. The radius of the circle passing through the vertices of the triangle ABC , is



- (A) $\frac{8\sqrt{15}}{5}$ (B) $\frac{3\sqrt{15}}{5}$ (C) $3\sqrt{5}$ (D) $3\sqrt{2}$

MCR007

8. $(6, 0)$, $(0, 6)$ and $(7, 7)$ are the vertices of a triangle. The circle inscribed in the triangle has the equation
 (A) $x^2 + y^2 - 9x + 9y + 36 = 0$ (B) $x^2 + y^2 - 9x - 9y + 36 = 0$
 (C) $x^2 + y^2 + 9x - 9y + 36 = 0$ (D) $x^2 + y^2 - 9x - 9y - 36 = 0$ **MCR008**
9. The line joining $(5, 0)$ to $(10\cos\theta, 10\sin\theta)$ is divided internally in the ratio $2 : 3$ at P . If θ varies then the locus of P is :
 (A) a pair of straight lines (B) a circle
 (C) a straight line (D) a second degree curve which is not a circle **MCR009**
10. The locus of the center of the circles such that the point $(2, 3)$ is the mid point of the chord $5x + 2y = 16$ is
 (A) $2x - 5y + 11 = 0$ (B) $2x + 5y - 11 = 0$
 (C) $2x + 5y + 11 = 0$ (D) none **MCR010**
11. In the xy -plane, the length of the shortest path from $(0, 0)$ to $(12, 16)$ that does not go inside the circle $(x - 6)^2 + (y - 8)^2 = 25$ is
 (A) $10\sqrt{3}$ (B) $10\sqrt{5}$ (C) $10\sqrt{3} + \frac{5\pi}{3}$ (D) $10 + 5\pi$ **MCR139**
12. Tangents PA and PB are drawn to the circle $x^2 + y^2 = 4$, then the locus of the point P if the triangle PAB is equilateral, is equal to-
 (A) $x^2 + y^2 = 16$ (B) $x^2 + y^2 = 8$ (C) $x^2 + y^2 = 64$ (D) $x^2 + y^2 = 32$ **MCR011**
13. In the xy plane, the segment with end points $(3, 8)$ and $(-5, 2)$ is the diameter of the circle. The point $(k, 10)$ lies on the circle for
 (A) no value of k (B) exactly one integral k
 (C) exactly one non integral k (D) two real values of k **MCR012**
14. The points (x_1, y_1) , (x_2, y_2) , (x_1, y_2) and (x_2, y_1) are always
 (A) collinear (B) concyclic
 (C) vertices of a square (D) vertices of a rhombus **MCR140**
15. Locus of all point $P(x, y)$ satisfying $x^3 + y^3 + 3xy = 1$ consists of union of
 (A) a line and an isolated point (B) a line pair and an isolated point
 (C) a line and a circle (D) a circle and a isolated point. **MCR141**
16. Consider the points $P(2, 1)$; $Q(0, 0)$; $R(4, -3)$ and the circle $S: x^2 + y^2 - 5x + 2y - 5 = 0$
 (A) exactly one point lies outside S (B) exactly two points lie outside S
 (C) all the three points lie outside S (D) none of the point lies outside S **MCR013**

17. The angle between the two tangents from the origin to the circle $(x - 7)^2 + (y + 1)^2 = 25$ equals
 (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$ **MCR014**
18. The area of the quadrilateral formed by the tangents from the point $(4, 5)$ to the circle $x^2 + y^2 - 4x - 2y - 11 = 0$ with the pair of radii through the points of contact of the tangents is:
 (A) 4 sq. units (B) 8 sq. units (C) 6 sq. units (D) none **MCR015**
19. If a circle of constant radius $3k$ passes through the origin 'O' and meets co-ordinate axes at A and B then the locus of the centroid of the triangle OAB is -
 (A) $x^2 + y^2 = (2k)^2$ (B) $x^2 + y^2 = (3k)^2$ (C) $x^2 + y^2 = (4k)^2$ (D) $x^2 + y^2 = (6k)^2$ **MCR142**
20. If L_1 and L_2 are the length of the tangent from $(0, 5)$ to the circles $x^2 + y^2 + 2x - 4 = 0$ and $x^2 + y^2 - 2x - y + 1 = 0$ then
 (A) $L_1 = 2L_2$ (B) $L_2 = 2L_1$ (C) $L_1 = L_2$ (D) $L_1^2 = L_2$ **MCR016**
21. From $(3, 4)$ chords are drawn to the circle $x^2 + y^2 - 4x = 0$. The locus of the mid points of the chords is:
 (A) $x^2 + y^2 - 5x - 4y + 6 = 0$ (B) $x^2 + y^2 + 5x - 4y + 6 = 0$
 (C) $x^2 + y^2 - 5x + 4y + 6 = 0$ (D) $x^2 + y^2 - 5x - 4y - 6 = 0$ **MCR017**
22. B and C are fixed points having co-ordinates $(3, 0)$ and $(-3, 0)$ respectively. If the vertical angle BAC is 90° , then the locus of the centroid of the $\triangle ABC$ has the equation:
 (A) $x^2 + y^2 = 1$ (B) $x^2 + y^2 = 2$ (C) $9(x^2 + y^2) = 1$ (D) $9(x^2 + y^2) = 4$ **MCR143**
23. Combined equation to the pair of tangents drawn from the origin to the circle $x^2 + y^2 + 4x + 6y + 9 = 0$ is
 (A) $3(x^2 + y^2) = (x + 2y)^2$ (B) $2(x^2 + y^2) = (3x + y)^2$
 (C) $9(x^2 + y^2) = (2x + 3y)^2$ (D) $x^2 + y^2 = (2x + 3y)^2$ **MCR018**
24. A circle of radius 5 is tangent to the line $4x - 3y = 18$ at $M(3, -2)$ and lies above the line. The equation of the circle, is-
 (A) $x^2 + y^2 - 6x + 4y - 12 = 0$ (B) $x^2 + y^2 + 2x - 2y - 3 = 0$
 (C) $x^2 + y^2 + 2x - 2y - 23 = 0$ (D) $x^2 + y^2 + 6x + 4y - 12 = 0$ **MCR019**
25. Tangents are drawn from $(4, 4)$ to the circle $x^2 + y^2 - 2x - 2y - 7 = 0$ to meet the circle at A and B. The length of the chord AB is
 (A) $2\sqrt{3}$ (B) $3\sqrt{2}$ (C) $2\sqrt{6}$ (D) $6\sqrt{2}$ **MCR144**
26. Tangents are drawn to a unit circle with centre at the origin from each point on the line $2x + y = 4$. Then the equation to the locus of the middle point of the chord of contact is -
 (A) $2(x^2 + y^2) = x + y$ (B) $2(x^2 + y^2) = x + 2y$ (C) $4(x^2 + y^2) = 2x + y$ (D) none **MCR145**

27. Chord AB of the circle $x^2 + y^2 = 100$ passes through the point $(7, 1)$ and subtends an angle of 60° at the circumference of the circle. If m_1 and m_2 are the slopes of two such chords then the value of $m_1 m_2$, is

(A) -1 (B) 1 (C) $7/12$ (D) -3

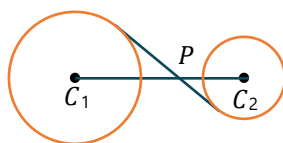
MCR146

28. Sum of the abscissa and ordinate of the centre of the circle touching the line $3x + y + 2 = 0$ at the point $(-1, 1)$ and passing through the point $(3, 5)$ is-

(A) 2 (B) 3 (C) 4 (D) 5

MCR147

29. In the figure given, two circles with centres C_1 and C_2 are 35 units apart, i.e. $C_1 C_2 = 35$. The radii of the circles with centres C_1 and C_2 are 12 and 9 respectively. If P is the intersection of $C_1 C_2$ and a common internal tangent to the circles, then $l(C_1 P)$ equals-



(A) 18 (B) 20 (C) 12 (D) 15

MCR148

30. Let C_1 and C_2 are circles defined by $x^2 + y^2 - 20x + 64 = 0$ and $x^2 + y^2 + 30x + 144 = 0$. The length of the shortest line segment PQ that is TANGENT to C_1 at P and to C_2 at Q is -

(A) 15 (B) 18 (C) 20 (D) 24

MCR020

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. $\frac{x-x_1}{\cos \theta} = \frac{y-y_1}{\sin \theta} = r$, represents: (Where x_1, y_1 are constant)
 (A) equation of a straight line, if θ is constant and r is variable
 (B) equation of a circle, if r is constant and θ is a variable
 (C) a straight line passing through a fixed point and having a known slope
 (D) a circle with a known centre and a given radius. MCR049
2. Which of the following lines have the intercepts of equal lengths on the circle, $x^2 + y^2 - 2x + 4y = 0$?
 (A) $3x - y = 0$ (B) $x + 3y = 0$
 (C) $x + 3y + 10 = 0$ (D) $3x - y - 10 = 0$ MCR050
3. One of the diameter of the circle circumscribing the rectangle $ABCD$ is $x - 3y + 1 = 0$. If two vertices of rectangle are the points $(-2, 5)$ and $(6, 5)$ respectively, then which of the following hold(s) good?
 (A) Area of rectangle $ABCD$ is 64 square units.
 (B) Centre of circle is $(2, 1)$
 (C) The other two vertices of the rectangle are $(-2, -3)$ and $(6, -3)$
 (D) Equation of sides are $x = -2, y = -3, x = 5$ and $y = 6$. MCR051
4. Three concentric circles of which the biggest is $x^2 + y^2 = 1$, have their radii in A.P. If the line $y = x + 1$ cuts all the circles in real and distinct points. The permissible values of common difference of A.P. is/are
 (A) 0.4 (B) 0.6 (C) 0.01 (D) 0.1 MCR052
5. The equation of circles passing through $(3, -6)$ touching both the axes is
 (A) $x^2 + y^2 - 6x + 6y + 9 = 0$ (B) $x^2 + y^2 + 6x - 6y + 9 = 0$
 (C) $x^2 + y^2 + 30x - 30y + 225 = 0$ (D) $x^2 + y^2 - 30x + 30y + 225 = 0$ MCR053
6. Tangents are drawn to the circle $x^2 + y^2 = 50$ from a point 'P' lying on the x -axis. These tangents meet the y -axis at points ' P_1 ' and ' P_2 '. Possible coordinates of 'P' so that area of triangle PP_1P_2 is minimum, is/are
 (A) $(10, 0)$ (B) $(10\sqrt{2}, 0)$ (C) $(-10, 0)$ (D) $(-10\sqrt{2}, 0)$ MCR054
7. The equations of the tangents drawn from the origin to the circle, $x^2 + y^2 - 2rx - 2hy + h^2 = 0$ are :
 (A) $x = 0$ (B) $y = 0$
 (C) $(h^2 - r^2)x - 2rhy = 0$ (D) $(h^2 - r^2)x + 2rhy = 0$ MCR055

8. Which of the following is/are True?

The circles $x^2 + y^2 - 6x - 6y + 9 = 0$ and $x^2 + y^2 + 6x + 6y + 9 = 0$ are such that -

- (A) they do not intersect
- (B) they touch each other
- (C) their exterior common tangents are parallel.
- (D) their interior common tangents are perpendicular.

MCR056

9. For the circles $x^2 + y^2 - 10x + 16y + 89 - r^2 = 0$ and $x^2 + y^2 + 6x - 14y + 42 = 0$ which of the following is/are true.

- (A) Number of integral values of r are 7 for which circles are intersecting.
- (B) Number of integral values of r are 9 for which circles are intersecting.
- (C) For r equal to 13 number of common tangents are 3.
- (D) For r equal to 21 number of common tangents are 2.

MCR057

10. Which of the following statement(s) is/are correct with respect to the circles $S_1 \equiv x^2 + y^2 - 4 = 0$ and $S_2 \equiv x^2 + y^2 - 2x - 4y + 4 = 0$?

- (A) S_1 and S_2 intersect at an angle of 90° .
- (B) The point of intersection of the two circle are $(2, 0)$ and $\left(\frac{6}{5}, \frac{8}{5}\right)$.
- (C) Length of the common of chord of S_1 and S_2 is $\frac{4}{\sqrt{5}}$.
- (D) The point $(2, 3)$ lies outside the circles S_1 and S_2 .

MCR058

11. Two circles of radii r_1 and r_2 are both touching the coordinate axes and intersecting each other orthogonally. The value of r_1/r_2 equals -

- (A) $2 + \sqrt{3}$
- (B) $\sqrt{3} + 1$
- (C) $2 - \sqrt{3}$
- (D) $2 + \sqrt{5}$

MCR149

12. Two circles $x^2 + y^2 + px + py - 7 = 0$ and $x^2 + y^2 - 10x + 2py + 1 = 0$ intersect each other orthogonally then the value of p is -

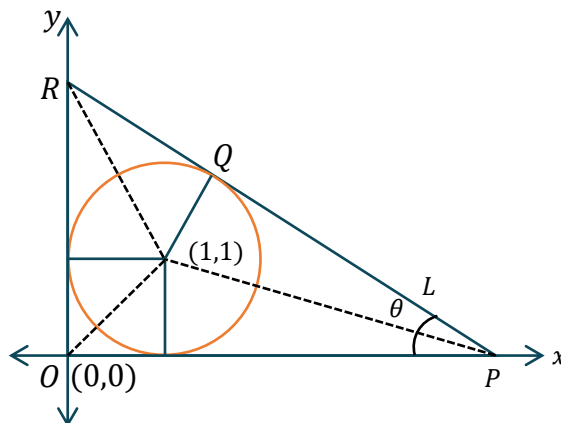
- (A) 1
- (B) 2
- (C) 3
- (D) 5

MCR150

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 13 to 15

In the diagram as shown, a circle is drawn with centre $C(1, 1)$ and radius 1 and a line L . The line L is tangential to the circle at Q . Further L meet the y –axis at R and the x –axis at P in such a way that the angle OPQ equals θ where $0 < \theta < \frac{\pi}{2}$.



- 13.** The coordinates of Q are
- (A) $(1 + \cos\theta, 1 + \sin\theta)$ (B) $(\sin\theta, \cos\theta)$
(C) $(1 + \sin\theta, \cos\theta)$ (D) $(1 + \sin\theta, 1 + \cos\theta)$
- 14.** Equation of the line PR is
- (A) $x\cos\theta + y\sin\theta = \sin\theta + \cos\theta + 1$ (B) $x\sin\theta + y\cos\theta = \cos\theta + \sin\theta - 1$
(C) $x\sin\theta + y\cos\theta = \cos\theta + \sin\theta + 1$ (D) $x\tan\theta + y = 1 + \cot\left(\frac{\theta}{2}\right)$

MCR059

- 15.** If the area bounded by the circle, the x -axis and PQ is $A(\theta)$, then $A\left(\frac{\pi}{4}\right)$ equals
- (A) $\sqrt{2} + 1 - \frac{3\pi}{8}$ (B) $\sqrt{2} - 1 + \frac{3\pi}{8}$ (C) $\sqrt{2} + 1 + \frac{\pi}{8}$ (D) $\sqrt{2} - 1 + \frac{\pi}{8}$

MCR060

MCR061

Paragraph for Question No. 16 to 18

Let S_1, S_2, S_3 be the circles $x^2 + y^2 + 3x + 2y + 1 = 0$, $x^2 + y^2 - x + 6y + 5 = 0$ and $x^2 + y^2 + 5x - 8y + 15 = 0$, respectively then

- 16.** Point from which length of tangents to these three circles is same is (α, β) then $(\alpha + \beta)$ is
(A) 4 (B) 5 (C) 6 (D) 7

MCR062

17. Equation of circle S_4 which cut orthogonally to all given circle is $x^2 + y^2 + ax + by + c = 0$ then $|a + b + c|$ is
- (A) 16 (B) 20 (C) 24 (D) 28

MCR063

18. Radical centre of circles S_1, S_2 , & S_4 is (p, q) then $|p + q|$ is
 (A) 2.20 (B) 3.20 (C) 1.20 (D) 4.20

MCR064

MATRIX MATCH TYPE QUESTION

19. **Column - I** **Column - II**
- | | |
|---|--------|
| (A) If director circle of two given circles C_1 and C_2 of equal radii touches each other, then ratio of length of internal common tangent of C_1 and C_2 to their radii equals to | (P) 13 |
| (B) Let two circles having radii r_1 and r_2 are orthogonal to each other. If length of their common chord is k times the square root of harmonic mean between squares of their radii, then k^4 equals to | (Q) 7 |
| (C) The axes are translated so that the new equation of the circle $x^2 + y^2 - 5x + 2y - 5 = 0$ has no first degree terms and the new equation $x^2 + y^2 = \frac{\lambda^2}{4}$, then a value of λ is | (R) 4 |
| (D) The number of integral points which lie on or inside the circle $x^2 + y^2 = 4$ is | (S) 2 |

MCR065

20. Consider two circles C_1 of radius ' a ' and C_2 of radius ' b ' ($b > a$) both lying in the first quadrant and touching the coordinate axes. In each of the conditions listed in **Column-I**, the ratio of b/a is given in **Column-II**.

- | Column-I | Column-II |
|---|---------------------|
| (A) C_1 and C_2 touch each other | (P) $2 + \sqrt{2}$ |
| (B) C_1 and C_2 are orthogonal | (Q) 3 |
| (C) C_1 and C_2 intersect so that the common chord is longest | (R) $2 + \sqrt{3}$ |
| (D) C_2 passes through the centre of C_1 | (S) $3 + 2\sqrt{2}$ |
| | (T) $3 - 2\sqrt{2}$ |

MCR066

EXERCISE (S)

1. A circle with center in the first quadrant is tangent to $y = x + 10$, $y = x - 6$, and the y -axis. Let (h, k) be the center of the circle. If the value of $(h + k) = a + b\sqrt{a}$ where $\sqrt{a}$ is a surd, find the value of $a + b$.
MCR151
2. If the points $(\lambda, -\lambda)$ lies inside the circle $x^2 + y^2 - 4x + 2y - 8 = 0$, then number of integers in the range of λ is
MCR021
3. Circles C_1 and C_2 are externally tangent and they are both internally tangent to the circle C_3 . The radii of C_1 and C_2 are 4 and 10, respectively and the centres of the three circles are collinear. A chord of C_3 is also a common internal tangent of C_1 and C_2 . Given that the length of the chord is $\frac{m\sqrt{n}}{p}$ where m, n and p are positive integers, m and p are relatively prime and n is not divisible by the square of any prime, find the value of $(m + n + p)$.
MCR022
4. One of the diameters of the circle circumscribing the rectangle $ABCD$ is $4y = x + 7$. If A & B are the points $(-3, 4)$ & $(5, 4)$ respectively, then find the area of the rectangle. (in square unit)
MCR152
5. Consider a circle S with centre at the origin and radius 4. Four circles A, B, C and D each with radius unity and centres $(-3, 0)$, $(-1, 0)$, $(1, 0)$ and $(3, 0)$ respectively are drawn. A chord PQ of the circle S touches the circle B and passes through the centre of the circle C . If the length of this chord can be expressed as $\sqrt{x}$, find x .
MCR023
6. A line with gradient 2 is passing through the point $P(1, 7)$ and touches the circle $x^2 + y^2 + 16x + 12y + c = 0$ at the point Q . If (a, b) are the coordinates of the point Q , then find the value of $(7a + 7b + c)$.
MCR024
7. Through a given point $P(5, 2)$, secants are drawn to cut the circle $x^2 + y^2 = 25$ at points $A_1(B_1)$, $A_2(B_2)$, $A_3(B_3)$, $A_4(B_4)$ and $A_5(B_5)$ such that $PA_1 + PB_1 = 5$, $PA_2 + PB_2 = 6$, $PA_3 + PB_3 = 7$, $PA_4 + PB_4 = 8$ and $PA_5 + PB_5 = 9$. Find the value of $\sum_{i=1}^5 PA_i^2 + \sum_{i=1}^5 PB_i^2$.
[Note : $A_r(B_r)$ denotes that the line passing through $P(5, 2)$ meets the circle $x^2 + y^2 = 25$ at two points A_r and B_r .]
MCR025
8. Real number x, y satisfies $x^2 + y^2 = 1$. If the maximum and minimum value of the expression are $z = \frac{4-y}{7-x}$ M and m respectively, then find the value $(2M + 6m)$.
MCR153

9. Let $S_1 = 0$ and $S_2 = 0$ be two circles intersecting at $P(6, 4)$ and both are tangent to x -axis and line $y = mx$ (where $m > 0$). If product of radii of the circles $S_1 = 0$ and $S_2 = 0$ is $\frac{52}{3}$, then the value of m^2 is
- MCR154**
10. If the circle $x^2 + y^2 + 4x + 22y + a = 0$ bisects the circumference of the circle $x^2 + y^2 - 2x + 8y - b = 0$ (where $a, b > 0$), then find the maximum value of (ab) .
- MCR155**

EXERCISE - JEE (Main) PYQ

1. Three circles of radii a, b, c ($a < b < c$) touch each other externally. If they have x -axis as a common tangent, then: [JEE (Main) 2019]

(1) $\frac{1}{\sqrt{a}} = \frac{1}{\sqrt{b}} + \frac{1}{\sqrt{c}}$

(2) a, b, c are in A.P.

(3) $\sqrt{a}, \sqrt{b}, \sqrt{c}$ are in A.P.

(4) $\frac{1}{\sqrt{b}} = \frac{1}{\sqrt{a}} + \frac{1}{\sqrt{c}}$

MCR026

2. If a circle C passing through the point $(4, 0)$ touches the circle $x^2 + y^2 + 4x - 6y = 12$ externally at the point $(1, -1)$, then the radius of C is: [JEE (Main) 2019]

(1) $\sqrt{57}$

(2) 4

(3) $2\sqrt{5}$

(4) 5

MCR027

3. If the area of an equilateral triangle inscribed in the circle, $x^2 + y^2 + 10x + 12y + c = 0$ is $27\sqrt{3}$ sq. units then c is equal to: [JEE (Main) 2019]

(1) 20

(2) 25

(3) 13

(4) -25

MCR028

4. A square is inscribed in the circle $x^2 + y^2 - 6x + 8y - 103 = 0$ with its sides parallel to the coordinate axes. Then the distance of the vertex of this square which is nearest to the origin is :- [JEE (Main) 2019]

(1) 13

(2) $\sqrt{137}$

(3) 6

(4) $\sqrt{41}$

MCR029

5. If a variable line, $3x + 4y - \lambda = 0$ is such that the two circles $x^2 + y^2 - 2x - 2y + 1 = 0$ and $x^2 + y^2 - 18x - 2y + 78 = 0$ are on its opposite sides, then the set of all values of λ is the interval :- [JEE (Main) 2019]

(1) $[12, 21]$

(2) $(2, 17)$

(3) $(23, 31)$

(4) $[13, 23]$

MCR031

6. The sum of the squares of the lengths of the chords intercepted on the circle, $x^2 + y^2 = 16$, by the lines, $x + y = n, n \in N$, where N is the set of all natural numbers, is: [JEE (Main) 2019]

(1) 320

(2) 160

(3) 105

(4) 210

MCR032

7. If a tangent to the circle $x^2 + y^2 = 1$ intersects the coordinate axes at distinct points P and Q , then the locus of the mid-point of PQ is [JEE (Main) 2019]

(1) $x^2 + y^2 - 2xy = 0$

(2) $x^2 + y^2 - 16x^2y^2 = 0$

(3) $x^2 + y^2 - 4x^2y^2 = 0$

(4) $x^2 + y^2 - 2x^2y^2 = 0$

MCR033

8. The common tangent to the circles $x^2 + y^2 = 4$ and $x^2 + y^2 + 6x + 8y - 24 = 0$ also passes through the point :- [JEE (Main) 2019]

(1) $(-4, 6)$

(2) $(6, -2)$

(3) $(-6, 4)$

(4) $(4, -2)$

MCR034

9. Let the tangents drawn from the origin to the circle, $x^2 + y^2 - 8x - 4y + 16 = 0$ touch it at the points A and B . The $(AB)^2$ is equal to: [JEE (Main) 2020]
 (1) $\frac{52}{5}$ (2) $\frac{32}{5}$ (3) $\frac{56}{5}$ (4) $\frac{64}{5}$ MCR035
10. If a line, $y = mx + c$ is a tangent to the circle, $(x - 3)^2 + y^2 = 1$ and it is perpendicular to a line L_1 , where L_1 is the tangent to the circle, $x^2 + y^2 = 1$ at the point $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$, then [JEE (Main) 2020]
 (1) $c^2 - 6c + 7 = 0$ (2) $c^2 + 6c + 7 = 0$
 (3) $c^2 + 7c + 6 = 0$ (4) $c^2 - 7c + 6 = 0$ MCR036
11. If the curves, $x^2 - 6x + y^2 + 8 = 0$ and $x^2 - 8y + y^2 + 16 - k = 0, (k > 0)$ touch each other at a point, then the largest value of k is _____. [JEE (Main) 2020]
MCR037
12. If one of the diameters of the circle $x^2 + y^2 - 2x - 6y + 6 = 0$ is a chord of another circle 'C', whose center is at $(2, 1)$, then its radius is _____. [JEE (Main) 2021]
MCR038
13. If the locus of the mid-point of the line segment from the point $(3, 2)$ to a point on the circle, $x^2 + y^2 = 1$ is a circle of radius r , then r is equal to : [JEE (Main) 2021]
 (1) 1 (2) $\frac{1}{2}$ (3) $\frac{1}{3}$ (4) $\frac{1}{4}$ MCR039
14. Let the normals at all the points on a given curve pass through a fixed point (a, b) . If the curve passes through $(3, -3)$ and $(4, -2\sqrt{2})$, and given that $a - 2\sqrt{2}b = 3$, then $(a^2 + b^2 + ab)$ is equal to. [JEE (Main) 2021]
MCR040
15. Let the lengths of intercepts on x -axis and y -axis made by the circle $x^2 + y^2 + ax + 2ay + c = 0$, ($a < 0$) be $2\sqrt{2}$ and $2\sqrt{5}$, respectively. Then the shortest distance from origin to a tangent to this circle which is perpendicular to the line $x + 2y = 0$, is equal to : [JEE (Main) 2021]
 (1) $\sqrt{11}$ (2) $\sqrt{7}$ (3) $\sqrt{6}$ (4) $\sqrt{10}$ MCR041
16. The minimum distance between any two points P_1 and P_2 while considering point P_1 on one circle and point P_2 on the other circle for the given circles' equations
 $x^2 + y^2 - 10x - 10y + 41 = 0$
 $x^2 + y^2 - 24x - 10y + 160 = 0$ is _____. [JEE (Main) 2021]
MCR042
17. Let a circle $C: (x - h)^2 + (y - k)^2 = r^2, k > 0$, touch the x -axis at $(1, 0)$. If the line $x + y = 0$ intersects the circle C at P and Q such that the length of the chord PQ is 2, then the value of $h + k + r$ is equal to. [JEE (Main) 2022]
MCR043

18. The set of values of k for which the circle $C: 4x^2 + 4y^2 - 12x + 8y + k = 0$ lies inside the fourth quadrant and the point $\left(1, -\frac{1}{3}\right)$ lies on or inside the circle C is : **[JEE (Main) 2022]**
- (1) An empty set (2) $\left[6, \frac{95}{9}\right]$ (3) $\left[\frac{80}{9}, 10\right]$ (4) $\left(9, \frac{92}{9}\right]$
- MCR044**
19. A rectangle R with end points of the one of its sides as $(1, 2)$ and $(3, 6)$ is inscribed in a circle. If the equation of a diameter of the circle is $2x - y + 4 = 0$, then the area of R is _____. **[JEE(Main)-2022]**
- MCR045**
20. Let the tangent to the circle $C_1: x^2 + y^2 = 2$ at the point $M(-1, 1)$ intersect the circle $C_2: (x - 3)^2 + (y - 2)^2 = 5$, at two distinct points A and B . If the tangents to C_2 at the points A and B intersect at N , then the area of the triangle ANB is equal to : **[JEE (Main) 2022]**
- (1) $\frac{1}{2}$ (2) $\frac{2}{3}$ (3) $\frac{1}{6}$ (4) $\frac{5}{3}$
- MCR046**
21. Let the abscissae of the two points P and Q on a circle be the roots of $x^2 - 4x - 6 = 0$ and the ordinates of P and Q be the roots of $y^2 + 2y - 7 = 0$. If PQ is a diameter of the circle $x^2 + y^2 + 2ax + 2by + c = 0$, then the value of $(a + b - c)$ is **[JEE (Main) 2022]**
- (1) 12 (2) 13 (3) 14 (4) 16
- MCR047**
22. Points $P(-3, 2)$, $Q(9, 10)$ and $R(\alpha, 4)$ lie on a circle C with PR as its diameter. The tangents to C at the points Q and R intersect at the point S . If S lies on the line $2x - ky = 1$, then k is equal to _____. **[JEE (Main) 2023]**
- MCR157**
23. The set of all values of a^2 for which the line $x + y = 0$ bisects two distinct chords drawn from a point $P\left(\frac{1+a}{2}, \frac{1-a}{2}\right)$ on the circle $2x^2 + 2y^2 - (1+a)x - (1-a)y = 0$ is equal to: **[JEE (Main) 2023]**
- (1) $(8, \infty)$ (2) $(4, \infty)$ (3) $[0, 4]$ (4) $(2, 12]$
- MCR158**
24. If the tangents at the points P and Q on the circle $x^2 + y^2 - 2x + y = 5$ meet at the point $R\left(\frac{9}{4}, 2\right)$, then the area of the triangle PQR is **[JEE (Main) 2023]**
- (1) $\frac{13}{4}$ (2) $\frac{13}{8}$ (3) $\frac{5}{4}$ (4) $\frac{5}{8}$
- MCR048**
25. Two circles in the first quadrant of radii r_1 and r_2 touch the coordinate axes. Each of them cuts off an intercept of 2 units with the line $x + y = 2$. Then $r_1^2 + r_2^2 - r_1 r_2$ is equal to _____. **[JEE (Main) 2023]**
- MCR159**

26. Let C be a circle with radius $\sqrt{10}$ units and centre at the origin. Let the line $x + y = 2$ intersects the circle C at the points P and Q . Let MN be a chord of C of length 2 unit and slope -1 . Then, a distance (in units) between the chord PQ and the chord MN is [JEE (Main) 2024]
 (1) $2 - \sqrt{3}$ (2) $3 - \sqrt{2}$ (3) $\sqrt{2} - 1$ (4) $\sqrt{2} + 1$
MCR160
27. A square is inscribed in the circle $x^2 + y^2 - 10x - 6y + 30 = 0$. One side of this square is parallel to $y = x + 3$. If (x_i, y_i) are the vertices of the square, then $\sum (x_i^2 + y_i^2)$ is equal to : [JEE (Main) 2024]
 (1) 148 (2) 156 (3) 160 (4) 152
MCR161
28. Let the circles $C_1 : (x - \alpha)^2 + (y - \beta)^2 = r_1^2$ and $C_2 : (x - 8)^2 + \left(y - \frac{15}{2}\right)^2 = r_2^2$ touch each other externally at the point $(6, 6)$. If the point $(6, 6)$ divides the line segment joining the centres of the circles C_1 and C_2 internally in the ratio $2 : 1$, then $(\alpha + \beta) + 4(r_1^2 + r_2^2)$ equals [JEE (Main) 2024]
 (1) 110 (2) 130 (3) 125 (4) 145
MCR162
29. Let the centre of a circle, passing through the point $(0, 0)$, $(1, 0)$ and touching the circle $x^2 + y^2 = 9$, be (h, k) . Then for all possible values of the coordinates of the centre (h, k) , $4(h^2 + k^2)$ is equal to. [JEE (Main) 2024]
MCR163
30. Consider two circles $C_1 : x^2 + y^2 = 25$ and $C_2 : (x - \alpha)^2 + y^2 = 16$, where $\alpha \in (5, 9)$. Let the angle between the two radii (one to each circle) drawn from one of the intersection points of C_1 and C_2 be $\sin^{-1}\left(\frac{\sqrt{63}}{8}\right)$. If the length of common chord of C_1 and C_2 is β , then the value of $(\alpha\beta)^2$ equals. [JEE (Main) 2024]
MCR164

EXERCISE - JEE (Advanced) PYQ

1. Circle(s) touching x -axis at a distance 3 from the origin and having an intercept of length $2\sqrt{7}$ on y -axis is (are) [JEE (Advanced) 2013]
 (A) $x^2 + y^2 - 6x + 8y + 9 = 0$ (B) $x^2 + y^2 - 6x + 7y + 9 = 0$
 (C) $x^2 + y^2 - 6x - 8y + 9 = 0$ (D) $x^2 + y^2 - 6x - 7y + 9 = 0$
MCR067
2. A circle S passes through the point $(0, 1)$ and is orthogonal to the circles $(x - 1)^2 + y^2 = 16$ and $x^2 + y^2 = 1$. Then:- [JEE (Advanced) 2014]
 (A) radius of S is 8 (B) radius of S is 7
 (C) centre of S is $(-7, 1)$ (D) centre of S is $(-8, 1)$
MCR068
3. Let RS be the diameter of the circle $x^2 + y^2 = 1$, where S is the point $(1, 0)$. Let P be a variable point (other than R and S) on the circle and tangents to the circle at S and P meet at the point Q . The normal to the circle at P intersects a line drawn through Q parallel to RS at point E . then the locus of E passes through the point(s)- [JEE (Advanced) 2016]
 (A) $\left(\frac{1}{3}, \frac{1}{\sqrt{3}}\right)$ (B) $\left(\frac{1}{4}, \frac{1}{2}\right)$
 (C) $\left(\frac{1}{3}, -\frac{1}{\sqrt{3}}\right)$ (D) $\left(\frac{1}{4}, -\frac{1}{2}\right)$
MCR069
4. For how many values of p , the circle $x^2 + y^2 + 2x + 4y - p = 0$ and the coordinate axes have exactly three common points? [JEE (Advanced) 2017]
MCR071

Paragraph (Q.5 & 6)

Let S be the circle in the xy -plane defined by the equation $x^2 + y^2 = 4$.

(There are two question based on Paragraph "X", the question given below is one of them)

5. Let E_1E_2 and F_1F_2 be the chord of S passing through the point $P_0(1, 1)$ and parallel to the x -axis and the y -axis, respectively. Let G_1G_2 be the chord of S passing through P_0 and having slope -1 . Let the tangents to S at E_1 and E_2 meet at E_3 , the tangents of S at F_1 and F_2 meet at F_3 , and the tangents to S at G_1 and G_2 meet at G_3 . Then, the points E_3, F_3 and G_3 lie on the curve [JEE (Advanced) 2018]
 (A) $x + y = 4$ (B) $(x - 4)^2 + (y - 4)^2 = 16$
 (C) $(x - 4)(y - 4) = 4$ (D) $xy = 4$
MCR072
6. Let P be a point on the circle S with both coordinates being positive. Let the tangent to S at P intersect the coordinate axes at the points M and N . Then, the mid-point of the line segment MN must lie on the curve - [JEE (Advanced) 2018]
 (A) $(x + y)^2 = 3xy$ (B) $x^{\frac{2}{3}} + y^{\frac{2}{3}} = 2^{\frac{4}{3}}$
 (C) $x^2 + y^2 = 2xy$ (D) $x^2 + y^2 = x^2y^2$
MCR073

7. Let T be the line passing through the points $P(-2, 7)$ and $Q(2, -5)$. Let F_1 be the set of all pairs of circles (S_1, S_2) such that T is tangents to S_1 at P and tangent to S_2 at Q , and also such that S_1 and S_2 touch each other at a point, say, M . Let E_1 be the set representing the locus of M as the pair (S_1, S_2) varies in F_1 . Let the set of all straight line segments joining a pair of distinct points of E_1 and passing through the point $R(1, 1)$ be F_2 . Let E_2 be the set of the mid-points of the line segments in the set F_2 . Then, which of the following statement(s) is (are) **TRUE**? [JEE (Advanced) 2018]

- (A) The point $(-2, 7)$ lies in E_1 (B) The point $\left(\frac{4}{5}, \frac{7}{5}\right)$ does **NOT** lie in E_2
 (C) The point $\left(\frac{1}{2}, 1\right)$ lies in E_2 (D) The point $\left(0, \frac{3}{2}\right)$ does **NOT** lie in E_1

MCR074

8. A line $y = mx + 1$ intersects the circle $(x - 3)^2 + (y + 2)^2 = 25$ at the points P and Q . If the midpoint of the line segment PQ has x -coordinate $-\frac{3}{5}$, then which one of the following options is correct? [JEE (Advanced) 2019]

- (A) $6 \leq m < 8$ (B) $2 \leq m < 4$ (C) $4 \leq m < 6$ (D) $-3 \leq m < -1$

MCR075

9. Let the point B be the reflection of the point $A(2, 3)$ with respect to the line $8x - 6y - 23 = 0$. Let Γ_A and Γ_B be circles of radii 2 and 1 with centres A and B respectively. Let T be a common tangent to the circles Γ_A and Γ_B such that both the circles are on the same side of T . If C is the point of intersection of T and the line passing through A and B , then the length of the line segment AC is [JEE (Advanced) 2019]

MCR076

For (Q.10 to 11) Answer the following by appropriately matching the lists based on the information given in the paragraph

Let the circles $C_1: x^2 + y^2 = 9$ and $C_2: (x - 3)^2 + (y - 4)^2 = 16$, intersect at the points X and Y . Suppose that another circle $C_3: (x - h)^2 + (y - k)^2 = r^2$ satisfies the following conditions:

- (i) centre of C_3 is collinear with the centres of C_1 and C_2
 (ii) C_1 and C_2 both lie inside C_3 , and
 (iii) C_3 touches C_1 at M and C_2 at N .

Let the line through X and Y intersect C_3 at Z and W , and let a common tangent of C_1 and C_3 be a tangent to the parabola $x^2 = 8\alpha y$.

There are some expression given in the List-I whose values are given in List-II below :

List-I		List-II	
(I)	$2h + k$	(P)	6
(II)	$\frac{\text{Length of } ZW}{\text{Length of } XY}$	(Q)	$\sqrt{6}$
(III)	$\frac{\text{Area of triangle } MZN}{\text{Area of triangle } ZMW}$	(R)	$\frac{5}{4}$
(IV)	α	(S)	$\frac{21}{5}$
		(T)	$2\sqrt{6}$
		(U)	$\frac{10}{3}$

10. Which of the following is the only INCORRECT combination? [JEE (Advanced) 2019]
 (A) (IV), (S) (B) (IV), (U)
 (C) (III), (R) (D) (I), (P)

MCR077

11. Which of the following is the only CORRECT combination? [JEE (Advanced) 2019]
 (A) (II), (T) (B) (I), (S)
 (C) (I), (U) (D) (II), (Q)

MCR078

12. Let O be the centre of the circle $x^2 + y^2 = r^2$, where $r > \frac{\sqrt{5}}{2}$. Suppose PQ is a chord of this circle and the equation of the line passing through P and Q is $2x + 4y = 5$. If the centre of the circumcircle of the triangle OPQ lies on the line $x + 2y = 4$, then the value of r is [JEE (Advanced) 2020]

MCR079

13. Consider a triangle Δ whose two sides lie on the x -axis and the line $x + y + 1 = 0$. If the orthocenter of Δ is $(1, 1)$, then the equation of the circle passing through the vertices of the triangle Δ is [JEE (Advanced) 2021]
 (A) $x^2 + y^2 - 3x + y = 0$ (B) $x^2 + y^2 + x + 3y = 0$
 (C) $x^2 + y^2 + 2y - 1 = 0$ (D) $x^2 + y^2 + x + y = 0$

MCR080

Paragraph (Q.14 & 15)

Let $M = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x^2 + y^2 \leq r^2\}$, where $r > 0$. Consider the geometric progression $a_n = \frac{1}{2^{n-1}}$, $n = 1, 2, 3, \dots$. Let $S_0 = 0$ and, for $n \geq 1$, let S_n denote the sum of the first n terms of this progression. For $n \geq 1$, let C_n denote the circle with center $(S_{n-1}, 0)$ and radius a_n , and D_n denote the circle with center (S_{n-1}, S_{n-1}) and radius a_n .

14. Consider M with $r = \frac{1025}{513}$. Let k be the number of all those circles C_n that are inside M . Let l be the maximum possible number of circles among these k circles such that no two circles intersect. Then [JEE (Advanced) 2021]
 (A) $k + 2l = 22$ (B) $2k + l = 26$
 (C) $2k + 3l = 34$ (D) $3k + 2l = 40$

MCR081

15. Consider M with $r = \frac{(2^{199} - 1)\sqrt{2}}{2^{198}}$. The number of all those circles D_n that are inside M is [JEE (Advanced) 2021]
 (A) 198 (B) 199 (C) 200 (D) 201

MCR082

16. Let ABC be the triangle with $AB = 1, AC = 3$ and $\angle BAC = \frac{\pi}{2}$. If a circle of radius $r > 0$ touches the sides AB, AC and also touches internally the circumcircle of the triangle ABC , then the value of r is _____. [JEE (Advanced) 2022]

MCR083

17. Let G be a circle of radius $R > 0$. Let $G_1, G_2, \dots, G_n$ be n circles of equal radius $r > 0$. Suppose each of the n circles $G_1, G_2, \dots, G_n$ touches the circle G externally. Also, for $i = 1, 2, \dots, n-1$, the circle G_i touches G_{i+1} externally, and G_n touches G_1 externally. Then, which of the following statements is/are TRUE? [JEE (Advanced) 2022]

- (A) If $n = 4$, then $(\sqrt{2} - 1)r < R$ (B) If $n = 5$, then $r < R$
 (C) If $n = 8$, then $(\sqrt{2} - 1)r < R$ (D) If $n = 12$, then $\sqrt{2}(\sqrt{3} + 1)r > R$

MCR084

18. Let C_1 be the circle of radius 1 with center at the origin. Let C_2 be the circle of radius r with center at the point $A = (4, 1)$, where $1 < r < 3$. Two distinct common tangents PQ and ST of C_1 and C_2 are drawn. The tangent PQ touches C_1 at P and C_2 at Q . The tangent ST touches C_1 at S and C_2 at T . Mid points of the line segments PQ and ST are joined to form a line which meets the x -axis at a point B . If $AB = \sqrt{5}$, then the value of r^2 is [JEE (Advanced) 2023]

MCR085

19. Let the straight line $y = 2x$ touch a circle with center $(0, \alpha)$, $\alpha > 0$, and radius r at a point A_1 . Let B_1 be the point on the circle such that the line segment A_1B_1 is a diameter of the circle. Let $\alpha + r = 5 + \sqrt{5}$. Match each entry in **List-I** to the correct entry in **List-II**.

List-I		List-II	
(P)	α equals	(1)	$(-2, 4)$
(Q)	r equals	(2)	$\sqrt{5}$
(R)	A_1 equals	(3)	$(-2, 6)$
(S)	B_1 equals	(4)	5
		(5)	$(2, 4)$

The correct option is

- (A) $(P) \rightarrow (4)$ $(Q) \rightarrow (2)$ $(R) \rightarrow (1)$ $(S) \rightarrow (3)$
 (B) $(P) \rightarrow (2)$ $(Q) \rightarrow (4)$ $(R) \rightarrow (1)$ $(S) \rightarrow (3)$
 (C) $(P) \rightarrow (4)$ $(Q) \rightarrow (2)$ $(R) \rightarrow (5)$ $(S) \rightarrow (3)$
 (D) $(P) \rightarrow (2)$ $(Q) \rightarrow (4)$ $(R) \rightarrow (3)$ $(S) \rightarrow (5)$

[JEE (Advanced) 2024]

MCR165

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	A	A	B	D	A	B	B	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	A	B	B	A	D	C	B	A	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	A	C	C	B	C	A	C	B	C

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10	
Ans.	A,B	A,B,C,D	A,B,C	C,D	A,D	A,C	A,C	A,C,D	A,C	A,C,D	
Que.	11	12	13	14	15	16	17	18			
Ans.	A,C	B,C	D	C	A	B	C	A			
Que.	19			20							
Ans.	A→S;B→R;C→Q;D→P			A→S;B→R;C→Q;D→P							

EXERCISE (S)

1.	10	2.	4	3.	19	4.	32	5.	63
6.	4	7.	215	8.	4.00	9.	3.00	10.	625.00

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	4	2	4	1	4	3	2	4	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	36	3	2	9	3	1	7	4	16	3
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	1	3	1	4	7	2	4	2	9	1575

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,C	B,C	A,C	2.00	A	D	B,D	B	10.00	A
Que.	11	12	13	14	15	16		17	18	19
Ans.	D	2.00	B	D	B	0.83 or 0.84		C,D	2.00	C

Important Notes